



Technical Report - Commands & Configurations

AI-Powered Intrusion Detection System for Cloud Environments

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1. Track A Target Services on VM1

All commands in this chapter are executed on VM1 via SSH from the host:

```
ssh akrid@192.168.100.46
```

1.1 Install Services

```
sudo apt install -y apache2 openssh-server vsftpd
```

1.2 Enable and Start on Boot

```
sudo systemctl enable --now apache2
sudo systemctl enable --now ssh
sudo systemctl enable --now vsftpd
```

1.3 Verify All Three Running

```
sudo systemctl status apache2 ssh vsftpd --no-pager
```

Expected: active (running) for all three services.

1.4 Check Listening Ports

```
ss -tlnp | grep -E '80|22|21'
```

LISTEN	0	511	0.0.0.0:80	apache2
LISTEN	0	128	0.0.0.0:22	sshd
LISTEN	0	32	0.0.0.0:21	vsftpd

2. Track A Verification from VM2 (Kali)

All commands in this chapter are executed in the Kali terminal (VM2).

2.1 Test HTTP Apache

```
curl -I http://192.168.100.46
```

```
HTTP/1.1 200 OK
Date: Mon, 27 Apr 2026 10:48:38 GMT
Server: Apache/2.4.52 (Ubuntu)
Last-Modified: Mon, 27 Apr 2026 10:14:07 GMT
Content-Length: 10671
Content-Type: text/html
```

2.2 Test SSH Port 22

```
nc -zv 192.168.100.46 22
```

```
(UNKNOWN) [192.168.100.46] 22 (ssh) open
```

Note on inverse host lookup warning

The message `inverse host lookup failed: Unknown host` is harmless. It indicates the absence of a DNS reverse record for the private lab IP, which is expected. The critical result is `open`.

2.3 Test FTP Port 21

```
nc -zv 192.168.100.46 21
```

```
(UNKNOWN) [192.168.100.46] 21 (ftp) open
```

3. Track B Attack Tools Installation on VM2

3.1 Install kali-tools-top10 and hping3

```
sudo apt update && sudo apt install -y kali-tools-top10 hping3
```

3.2 Verify Tool Versions

```
nmap --version  
hydra --version  
hping3 --version  
msfconsole --version
```

```
Nmap version 7.99 ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu
```

```
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak
```

```
hping3 version 3.0.0-alpha-2
```

```
Framework Version: 6.4.126-dev
```

4. Track B Attack Tests

4.1 Attack Test 1 SSH Brute Force with Hydra

4.1.1 Create Password Wordlist

```
cat > /tmp/passwords.txt << EOF
password
123456
admin
root
toor
test
akrid
EOF
```

4.1.2 Run Hydra SSH Brute Force

```
hydra -l akrid -P /tmp/passwords.txt ssh://192.168.100.46 -t 4 -v
```

```
Hydra v9.6 starting at 2026-04-27 13:53:21
[DATA] max 4 tasks per 1 server, overall 4 tasks, 7 login tries
[DATA] attacking ssh://192.168.100.46:22/
[VERBOSE] Resolving addresses ... done
[INFO] Testing if password authentication is supported
[INFO] Successful, password authentication is supported
[STATUS] attack finished for 192.168.100.46
1 of 1 target completed, 0 valid password found
Hydra finished at 2026-04-27 13:53:29
```

Expected result

No valid password found is the correct result the wordlist was intentionally short and did not contain the real password. The attack generated real TCP traffic against SSH which is what the IDS needs to capture.

4.2 Attack Test 2 SYN Flood with hping3

```
sudo hping3 -S --flood -V -p 80 192.168.100.46
# Run for ~10 seconds then Ctrl+C
```

```
using eth0, addr: 192.168.100.28, MTU: 1500
HPING 192.168.100.46 (eth0 192.168.100.46): S set, 40 headers +
    0 data bytes
hping in flood mode, no replies will be shown
^C
192.168.100.46 hping statistic
5851174 packets transmitted, 0 packets received, 100% packet
    loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

4.3 Traffic Capture on VM1 During Attacks

Executed simultaneously on VM1 while attacks were running from VM2:

```
sudo tcpdump -i enp1s0 -n | grep "192.168.100.28" > logs.txt
```

4.3.1 SSH Brute Force Capture (head of logs.txt)

```
12:53:21.936474 IP 192.168.100.28.44578 > 192.168.100.46.22:
    Flags [S], seq 1527633744, win 64240, length 0
12:53:21.936562 IP 192.168.100.46.22 > 192.168.100.28.44578:
    Flags [S.], seq 3476509337, ack 1527633745, win 65160, length
    0
12:53:21.936797 IP 192.168.100.28.44578 > 192.168.100.46.22:
    Flags [.], ack 1, win 126, length 0
12:53:21.936797 IP 192.168.100.28.44578 > 192.168.100.46.22:
    Flags [P.], seq 1:24, length 23: SSH: SSH-2.0-libssh_0.11.3
12:53:21.943490 IP 192.168.100.46.22 > 192.168.100.28.44578:
    Flags [P.], seq 1:43, length 42: SSH: SSH-2.0-OpenSSH_8.9p1
    Ubuntu
```

4.3.2 SYN Flood Capture (tail of logs.txt)

```
12:54:17.140388 IP 192.168.100.46.80 > 192.168.100.28.20898:
    Flags [S.], seq 2634942518, ack 977707707, win 64240, length 0
12:54:17.140391 IP 192.168.100.28.20899 > 192.168.100.46.80:
    Flags [S], seq 614210212, win 512, length 0
12:54:17.140395 IP 192.168.100.46.80 > 192.168.100.28.20899:
    Flags [S.], seq 2302669792, ack 614210213, win 64240, length 0
12:54:17.140399 IP 192.168.100.28.20900 > 192.168.100.46.80:
    Flags [S], seq 97688348, win 512, length 0
12:54:17.140399 IP 192.168.100.28.20896 > 192.168.100.46.80:
    Flags [R], seq 1522412663, win 0, length 0
```

5. Track C Dataset Acquisition

5.1 Install Kaggle CLI

Executed inside the project virtual environment on the host:

```
source /mnt/lab-project/ids-project/venv/bin/activate
pip install kaggle
```

5.2 Configure Kaggle API Token

```
mkdir -p ~/.config/kaggle
mv ~/Downloads/kaggle.json ~/.config/kaggle/
chmod 600 ~/.config/kaggle/kaggle.json
```

5.3 Download Dataset

```
cd /mnt/lab-project/ids-project/datasets/cicids2017
kaggle datasets download -d chethuhn/network-intrusion-dataset
unzip network-intrusion-dataset.zip
rm network-intrusion-dataset.zip
```

5.4 Remove Unused Files

```
rm Thursday-WorkingHours-Afternoon-Infiltration.pcap_ISCX.csv
rm Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv
```

5.5 Verify Final Dataset Contents

```
ls -lh *.csv
```

```
Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv
Friday-WorkingHours-Morning.pcap_ISCX.csv
Monday-WorkingHours.pcap_ISCX.csv
Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv
Tuesday-WorkingHours.pcap_ISCX.csv
Wednesday-workingHours.pcap_ISCX.csv
```

6. ML Environment Setup

6.1 Create and Activate Virtual Environment

```
python3 -m venv venv
source venv/bin/activate
which python3
```

```
/mnt/lab-project/ids-project/venv/bin/python3
```

6.2 Install ML Frameworks

```
pip install --upgrade pip
pip install pandas numpy scikit-learn matplotlib seaborn
pip install mlflow
pip install shap
pip install torch torchvision \
    --index-url https://download.pytorch.org/whl/cu121
pip install kafka-python jupyterlab
```

6.3 Save Dependencies

```
pip freeze > requirements.txt
```

7. Storage Architecture Final Partition

7.1 Discover Unallocated Space

```
sudo parted /dev/nvme0n1 print free
```

Number	Start	End	Size	File system	Name
	17.4kB	1049kB	1031kB	Free Space	
1	1049kB	316MB	315MB	fat32	EFI system
	partition				
3	450MB	226GB	225GB	ntfs	Basic data
	partition				
4	226GB	258GB	31.7GB	ntfs	Basic data
	partition				
	258GB	295GB	37.7GB	Free Space	
9	295GB	381GB	86.1GB	ext4	primary
7	381GB	382GB	1049MB	fat32	
8	382GB	489GB	106GB	ext4	

7.2 Create New ext4 Partition

```
sudo parted /dev/nvme0n1
(parted) mkpart primary ext4 258GB 295GB
(parted) quit

sudo partprobe /dev/nvme0n1
sudo mkfs.ext4 /dev/nvme0n1p10 -L "lab-project"
```

7.3 Mount Permanently

```
sudo mkdir /mnt/lab-project
echo "LABEL=lab-project /mnt/lab-project ext4 defaults 0 2" | \
    sudo tee -a /etc/fstab
sudo mount -a
```

7.4 Final Storage Verification

```
df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/nvme0n1p8	97G	37G	56G	40%	/
/dev/nvme0n1p10	35G	7.1G	26G	22%	/mnt/lab-project
/dev/nvme0n1p9	79G	17G	58G	23%	/mnt/lab